Cloud Security Alliance Submission to the AI Action Plan RFI

Email to: [redacted]

Subject: AI Action Plan - Cloud Security Alliance Submission

Date: March 15th, 2025

Cloud Security Alliance (CSA) Response to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

AGENCY: Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation (NSF).

SUBMISSION BY: Cloud Security Alliance (CSA)

STATEMENT OF PUBLIC DISSEMINATION: This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Introduction

The Cloud Security Alliance (CSA) is a global leader in defining best practices for securing cloud computing and emerging technologies, including artificial intelligence (AI). As an industry-driven nonprofit, CSA collaborates with public and private sector stakeholders to establish security frameworks, research AI risk management, and support policy development that ensures AI innovation is secure, resilient, and fosters trust. For over 15 years, CSA has enabled early adoption of technologies by providing best practices and assurance methodologies for both solution providers and those organizations implementing these technologies. We welcome the opportunity to provide input on the AI Action Plan and recommend key priority policy actions to maintain the United States' leadership in AI while ensuring security and trustworthiness in its deployment.

Notable initiatives:

AI Safety Initiative (https://cloudsecurityalliance.org/ai-safety-initiative)
CSA's AI Safety Initiative is the premier coalition of trusted experts who converge to develop and deliver essential AI guidance and tools that empower organizations of all sizes to deploy AI solutions that are safe, responsible, and compliant.

Security, Trust, Assurance, and Risk (STAR) Program
(https://cloudsecurityalliance.org/star)
The STAR Registry is a free, publicly accessible registry that documents the security and privacy controls provided by popular cloud computing offerings, allowing public users to review and service providers to share security practices. STAR encompasses the key principles of transparency, rigorous auditing, and harmonization of standards outlined in the Cloud Controls Matrix (CCM) to reduce complexity and show security and compliance posture across regulations, standards, and frameworks.

Zero Trust Advancement Center (https://cloudsecurityalliance.org/zt)
Creating an industry "north star" for Zero Trust has huge implications in raising the cybersecurity baseline across the board and eliminating significant systemic risk. CSA's mission is to create research, training, professional credentialing and provide an online center for additional curated Zero Trust resources to understand and implement Zero Trust principles into business planning, enterprise architectures and technology deployments.

Key Recommendations for the AI Action Plan

1. Strengthening Cloud AI Infrastructure Security

AI in Cloud and Data Center Infrastructure.

- AI models rely on cloud-scale infrastructure, creating new security risks. The U.S. must lead in defining best practices for **AI model protection**, **secure deployment**, and **data integrity** to prevent adversarial exploits.
- Promote resilient cloud security architectures to support AI workloads while mitigating risks related to unauthorized access, data breaches, and AI model leakage.
- Address AI model security risks in cloud environments, including model theft, adversarial attacks, and unauthorized access.
- Establish best practices for AI security at hyperscale, ensuring scalability and resilience.

- Encourage industry collaboration on secure AI infrastructure, including security measures for hyperscale cloud environments.

Supporting AI Compute and Energy Efficiency.

- Invest in research on energy-efficient and secure AI computing, balancing innovation with sustainability and cybersecurity.
- Address growing concerns about AI energy consumption by incentivizing **efficient AI processing architectures**.
- Support research into sustainable AI infrastructure to optimize power usage without compromising security.

Mitigating AI Cloud Energy and Compute Constraints

- AI workloads are projected to **overwhelm cloud and energy resources** within the next few years. Strategic investments in **efficient AI computing**, **sustainable AI infrastructure**, and **decentralized AI processing** will be crucial to maintaining U.S. leadership.
- Anticipate increasing demand for AI-driven cloud computing and implement strategies to mitigate energy constraints and security risks.

Securing AI Compute Supply Chains

- The AI boom is straining compute resources and creating dependencies on foreign chip manufacturers. Policies should support **domestic AI compute security** and **protect critical semiconductor supply chains** from foreign influence and cyber threats.
- Strengthen supply chain security for AI accelerators (GPUs, TPUs) to prevent foreign dependencies and ensure national competitiveness.

2. Enable Secure AI Model Development and Agentic AI

AI Security and Risk Management

- The U.S. role as a global leader in AI depends on secure and trustworthy model adoption. National guidelines should focus on **securing AI training data**, preventing **model tampering**, and ensuring **verifiable AI system integrity**.
- Establish security best practices for **Agentic AI**—AI systems that operate with autonomy—ensuring their development and deployment adhere to strict security controls to prevent unintended consequences and adversarial manipulation.

- Assure AI model updates have been fully tested against **expected behaviors** prior to implementation into production and monitor for abnormalities in response

Zero Trust Strategy

- Apply **Zero Trust principles** to AI model development, ensuring that only verified, authorized entities can access and modify AI training data and models.
- Ensure the continuous authentication of all access and specified roles, especially for agentic behavior that may operate autonomously
- Support development of privacy-preserving AI techniques, such as differential privacy and federated learning, to protect sensitive data while enabling AI innovation.

AI Maturity Model

- Introduce an **AI Maturity Model** to assess organizational readiness and security posture for AI deployment, guiding enterprises on secure and responsible AI adoption.
- Support the adoption of AI explainability techniques that enhance confidence in AI-driven decision-making.
- Establish security guidelines for AI model development, covering threats such as data poisoning, adversarial attacks, and model inversion risks.
- Develop national guidelines for AI integrity, ensuring models are protected from manipulation and adversarial exploits.
- Support **transparent AI provenance tracking** to prevent data poisoning and unauthorized modifications.

Enhancing AI Privacy and Data Protection

- Strengthen security frameworks for sensitive training data in cloud AI systems.
- Expand privacy-preserving AI technologies such as **federated learning**, **homomorphic encryption, and confidential computing** to secure enterprise adoption.

3. Promoting Secure AI Model Assurance and Transparency

AI Cybersecurity Framework

- Adopt industry accepted AI cybersecurity controls frameworks, incorporating expanded cybersecurity best practices for model assurance, adversarial robustness, and secure AI supply chains.

- Develop voluntary assurance frameworks for AI systems, ensuring verifiable, interpretable, and trustworthy AI decision-making processes.
- Establish standardized AI transparency reporting, enabling enterprises and consumers to understand model behavior and reliability.
- Require transparency measures for AI system integrity, including secure model provenance and verifiable lineage.

Governance, Compliance, and Regulatory Coordination

- Establish harmonized AI security and governance policies that align with global frameworks while reducing regulatory burdens on AI innovation.
- Develop national security policies to protect AI supply chains from adversarial threats, including foreign influence on AI infrastructure and datasets.
- Harmonize industry-driven security standards, aligned with and encouraging the NIST AI Risk Management Framework and the CSA AI Controls Matrix.
- Encourage secure AI model development lifecycles with rigorous validation, red-teaming, and security testing.
- Define clear security and compliance requirements for AI models used in critical infrastructure and high-risk applications.

4. Advancing AI Cybersecurity to Protect National Interests

AI-Driven Defense Systems

- AI-powered cyber threats are evolving rapidly. The U.S. must develop **AI-driven defense systems** that enable real-time threat detection, adversarial AI mitigation, and resilient cloud-based security architectures.
- Implement **Zero Trust security models** for AI infrastructure to ensure continuous verification of AI systems, models, and data access.

AI-Driven Cybersecurity for Critical Infrastructure

- Prioritize **AI for cyber defense** to detect and mitigate threats at machine speed.
- Encourage government-industry collaboration on AI-driven security analytics to protect critical infrastructure.

5. AI Workforce Development and Training

AI Fundamentals and Expertise

- Expand AI security education programs to build a skilled workforce capable of addressing AI cybersecurity challenges.
- Invest in public-private partnerships to develop training initiatives focused on AI security best practices.
- Training and awareness for leveraging AI to support existing roles and responsibilities
- Establishing roles and responsibilities for advancing the use of AI

Strengthening International Collaboration and Competitiveness

- Engage in global partnerships to establish AI security norms and prevent fragmentation in AI security standards.
- Support AI security research collaborations between U.S. institutions and allied nations to advance AI trustworthiness and interoperability.
- Encourage responsible AI export policies that balance innovation with national security interests.

Conclusion

CSA strongly supports a balanced AI Action Plan that ensures robust security measures while fostering AI innovation. The importance of AI education, security frameworks, and assurance programs will enhance readiness and establish verifiable AI security practices to encourage AI adoption. Strengthening these areas will support sustainable AI adoption and resilience in evolving AI ecosystems. By incorporating security-by-design and Zero Trust principles, industry collaboration, and a risk-based approach, the United States can lead in AI development while ensuring safety and trust in AI technologies.

We appreciate the opportunity to contribute to this effort and welcome further engagement in shaping AI security policy.

Respectfully submitted,

Cloud Security Alliance